

Compact affine actions reduce to the linear-isometric case for hyperspaces of bounded convex sets

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Abstract

Jonard-Pérez asks whether $\text{CB}(L)$ is a G -equivariant absolute extensor when a compact group G acts continuously and affinely on a Banach space L and the induced action on $\text{CB}(L)$ is continuous. We give an affirmative answer. Haar-averaging any orbit produces a fixed point c . Translation by c conjugates the affine action to a continuous linear action, and the equivalent norm

$$\|x\|_G = \sup_{g \in G} \|gx\|$$

makes that action isometric. Equivalent norms induce bi-Lipschitz equivalent Hausdorff metrics on $\text{CB}(L)$, so the assumed hyperspace continuity is preserved. The problem is therefore exactly the linear-isometric case settled by Theorem 5.1 of the source paper.

1 Source question and available theorem

For a Banach space L , let $\text{CB}(L)$ be the hyperspace of all nonempty closed, bounded, convex subsets of L , equipped with the Hausdorff metric. The source is N. Jonard-Pérez, *Equivariant absolute extensor property on hyperspaces of convex sets*, arXiv:1401.2653v2 [1]. Its final Question 6.2 is reproduced below.

continuously on L by means of affinities. Then, the following sentences are true:

- (1) *Any closed and convex invariant subset $\mathcal{C}$ of $\text{cc}(L)$ is a G -AE. In particular $\text{cc}(L)$ is a G -AE.*
- (2) *If G is a Lie group, any convex invariant subset $\mathcal{C}$ of $\text{cc}(L)$ is a G -ANE. Additionally, if $\mathcal{C}$ has a fixed point, then $\mathcal{C} \in G\text{-AE}$.*

Final Remark. The map Φ defined in (6.1) can also be extended to an equivariant bijection from $\text{CB}(L)$ to $\text{CB}(\Phi(L))$. However, since the map Φ is not uniformly continuous, the hyperspaces $\text{CB}(L)$ and $\text{CB}(\Phi(L))$ may not be homeomorphic. For this reason, we cannot use the previous arguments to extend Theorem 6.1 to $\text{CB}(L)$, which leads to the following question.

Question 6.2. *Suppose that a compact group G acts continuously and affinely on a Banach space L in such a way that the induced action on $\text{CB}(L)$ is continuous. Is $\text{CB}(L) \in G\text{-AE}$?*

REFERENCES

1. S. A. Antonyan, *Retracts in categories of G -spaces*, Izvestiya Akademii Nauk Armyanskoi SSR. Matematika, Vol. 15, No. 5, (1980), 365-378.

The source had already proved the following linear-isometric result. In its terminology, a Banach G -space is a Banach space on which G acts linearly and the norm is G -invariant.

5. EQUIVARIANT ABSOLUTE EXTENSOR PROPERTY

In this section we prove the main results of this paper. Let us remark that if G is a Lie group, Theorem 5.1 is a particular case of [6 Corollary 4.6].

Theorem 5.1. *Let G be a compact group and L a Banach G -space. Then, $cc(L)$ is a G -AE. Additionally, if the induced action of G on $CB(L)$ is continuous, then $CB(L)$ is also a G -AE.*

Proof. As previously, let us denote by $\mathcal{K}$ the hyperspace $cc(L)$ or $CB(L)$ equipped with the induced (continuous) action of G . According to Theorem 4.1 $j(\mathcal{K})$ is an invariant closed and convex subset of a Banach G -space. Now, according to Theorem 2.1 $j(\mathcal{K})$ is a G -AE. Since $j(\mathcal{K})$ and $\mathcal{K}$ are G -homeomorphic, this directly implies both parts of the theorem. $\square$

We will say that a family $\mathcal{C} \subset \mathcal{K}$ is convex iff

$$\overline{tA + (1-t)B} \in \mathcal{C} \quad \text{for every } A, B \in \mathcal{C}, t \in [0, 1].$$

The following are examples of convex families:

- i) $CB(M)$ and $cc(M)$, where M is a closed convex subset of L .
- ii) The family of all infinite dimensional convex compacta of L .
- iii) The family of all finite dimensional convex compacta of L .

Thus the only issue is whether the compact affine action in Question 6.2 can be placed in that setting without changing the topology of $CB(L)$. It can.

2 Proof intuition

The source tries to replace the affine action by the orbit-map embedding $x \mapsto (g \mapsto gx)$ into $C(G, L)$. That map need not be uniformly continuous, which matters for Hausdorff hyperspaces of arbitrary bounded sets. A compact affine action admits a more direct reduction. Average one orbit against Haar probability measure. Its barycenter is fixed by the whole group. Once that point is declared to be the origin, the action is linear. Compactness then supplies a uniformly equivalent invariant norm. Unlike the orbit-map embedding, translation and equivalent renorming are bi-Lipschitz on the entire hyperspace of bounded sets.

3 Two reduction lemmas

Throughout, G is a compact topological group with normalized left Haar measure μ , and $T_g x = gx$ denotes the original action.

Lemma 1 (fixed point and linearization). *Let G act continuously and affinely on a real Banach space L . There is a point $c \in L$ fixed by G . The formula*

$$\rho_g y = T_g(c + y) - c$$

defines a continuous linear action of G on L , and translation $\tau(x) = x - c$ is equivariant from the original affine action to ρ .

Proof. Fix $x_0 \in L$. The orbit map $g \mapsto T_g x_0$ is continuous and has compact image. A compact subset of a Banach space is separable and bounded, so this map is Bochner integrable. Set

$$c = \int_G T_g x_0 d\mu(g).$$

For completeness, every continuous affine map $T : L \rightarrow L$ has the form $T(x) = Ax + b$ with A bounded and real-linear. Indeed, $F(x) = T(x) - T(0)$ fixes zero and preserves convex combinations. Comparing $F((x + y)/2)$ first with $(F(x) + F(y))/2$ and then with $F(x + y)/2$ proves additivity; homogeneity on $[0, 1]$, followed by additivity, gives real homogeneity. Continuity makes the resulting linear map bounded.

Consequently every T_h commutes with probability barycenters, and for $h \in G$,

$$\begin{aligned} T_h c &= \int_G T_h(T_g x_0) d\mu(g) \\ &= \int_G T_{hg} x_0 d\mu(g) = c, \end{aligned}$$

where the last equality is left invariance of μ . Hence c is fixed. Each map $y \mapsto T_g(c + y) - c$ is affine and fixes zero, so it is bounded and linear by the preceding observation. The action law and joint continuity are inherited from T , and

$$\tau(T_g x) = T_g x - c = \rho_g(x - c) = \rho_g \tau(x).$$

□

Lemma 2 (invariant renorming and hyperspaces). *For the action ρ from Lemma 1, define*

$$\|y\|_G = \sup_{g \in G} \|\rho_g y\|.$$

Then $\|\cdot\|_G$ is a complete, G -invariant norm equivalent to the original norm. If d_H and d_H^G are the Hausdorff metrics induced by $\|\cdot\|$ and $\|\cdot\|_G$, respectively, there is an $M < \infty$ such that, for all $A, B \in \text{CB}(L)$,

$$d_H(A, B) \leq d_H^G(A, B) \leq M d_H(A, B).$$

In particular, the two norm choices give the same underlying family $\text{CB}(L)$ and the same hyperspace topology.

Proof. For each $y \in L$, continuity of $g \mapsto \rho_g y$ and compactness of G make the displayed supremum finite. The family $\{\rho_g : g \in G\}$ is therefore pointwise bounded. The uniform boundedness principle gives

$$M := \sup_{g \in G} \|\rho_g\| < \infty.$$

Since the identity belongs to the family,

$$\|y\| \leq \|y\|_G \leq M \|y\|.$$

Thus the new norm is equivalent to the old one and hence complete. Moreover, for $h \in G$,

$$\|\rho_h y\|_G = \sup_{g \in G} \|\rho_{gh} y\| = \sup_{k \in G} \|\rho_k y\| = \|y\|_G,$$

so it is G -invariant.

Equivalent norms have the same closed and bounded subsets and preserve convexity. The pointwise norm inequalities imply the corresponding inequalities for distance to a set, hence for both directed excesses and therefore for the Hausdorff metrics. This proves the assertion. □

4 Affirmative answer

Theorem 3. *Suppose a compact group G acts continuously and affinely on a Banach space L . If the induced action on $\text{CB}(L)$ is continuous, then*

$$\text{CB}(L) \in G\text{-AE}.$$

Thus the answer to Question 6.2 of the source is yes.

Proof. Let c , ρ , and $\|\cdot\|_G$ be given by the two lemmas, and write $L_G = (L, \|\cdot\|_G, \rho)$. It is a Banach G -space in the exact sense of the source: the action is continuous and linear, and the norm is invariant.

Consider the induced map on hyperspaces

$$\Theta : \text{CB}(L) \longrightarrow \text{CB}(L_G), \quad \Theta(A) = A - c.$$

Translation preserves nonemptiness, closedness, boundedness, and convexity. It is an isometry for the original norm, while Lemma 2 shows that changing to the target Hausdorff metric is bi-Lipschitz. Hence Θ is a homeomorphism. It is equivariant because

$$\Theta(T_g A) = T_g A - c = \{T_g a - c : a \in A\} = \{\rho_g(a - c) : a \in A\} = \rho_g \Theta(A).$$

By hypothesis the induced G -action on the domain hyperspace is continuous. Conjugating by the G -homeomorphism Θ proves that the induced action on $\text{CB}(L_G)$ is continuous as well. Theorem 5.1 of the source now applies to the Banach G -space L_G and yields $\text{CB}(L_G) \in G\text{-AE}$. The $G\text{-AE}$ property is invariant under G -homeomorphism, so the same conclusion holds for the original $\text{CB}(L)$. $\square$

5 Verification, scope, and novelty notes

The continuity assumption in Question 6.2 has not been removed. It is not automatic: the source gives a linear-isometric compact-group action whose induced action on $\text{CB}(L)$ is discontinuous. The proof only shows that, when the assumed hyperspace action is continuous, passing from affine to linear coordinates and to an invariant equivalent norm does not destroy that continuity.

No metrizability of G is used. The only integration issue is harmless: although G itself may be nonmetrizable, the continuous orbit image is a compact subset of the metric space L , hence separable; the bounded orbit map is therefore Bochner integrable. All metric comparisons are global on bounded sets, rather than merely local on compacta.

A bounded novelty search on August 12, 2026 checked the four lightweight run indexes using the arXiv id, title, and the terms “affine action”, “ $\text{CB}(L)$ ”, “ $G\text{-AE}$ ”, “fixed point”, and “invariant norm”. External searches used the exact title, exact final question, arXiv id, and the core phrases above. They found the source paper but no later paper stating this reduction or answering Question 6.2. This supports, but does not establish, novelty; specialist literature review remains appropriate.

The packet is recommended for expert review as a likely valid full solution. The main points to audit are the Haar-barycenter fixed point, the uniform boundedness step in the invariant renorming, the bi-Lipschitz comparison of Hausdorff metrics, and the exact hypotheses of source Theorem 5.1.

References

- [1] N. Jonard-Pérez, *Equivariant absolute extensor property on hyperspaces of convex sets*, Topology and its Applications 177 (2014), 88–96; arXiv:1401.2653v2.